

Classification Methods on SpaceNet-7 Satellite Imagery

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Abstract

This study examines the feasibility of classifying seasonal patterns and identifying urban sectors within multi-temporal satellite imagery using a combination of supervised and unsupervised machine learning techniques. The dataset, derived from the SpaceNet-7 Multi-Temporal Urban Development Challenge, consists of 60 "Areas of Interest" (AOIs), each with monthly imagery over a two year period and corresponding geospatial building-footprint labels. The project uses image processing pipelines, including Rasterio ingestion, PCA reduction, and ResNet-18 embeddings, with classical classifiers, Gaussian Mixture Models (GMM) and Hidden Markov Models (HMM). Results demonstrate moderate success in supervised season classification, with strong performance varying significantly across AOIs. Unsupervised analyses reveal that latent seasonal states emerge only with looser seasonal definitions, or for AOIs with visually consistent phenology, whereas complex biome interactions limit global applicability. Sector segmentation experiments indicate that feature engineered GMM methods can reliably extract man-made vs. natural structures and further separate dense from sparse built environments. Brief, exploratory time series and building count predictor analyses were also performed and described.

1 Introduction

Urban development is one of the most visible and consequential indicators of human progress. Tracking the rate and patterns of urban change provides crucial insights for things like sustainable planning, infrastructure investment, and environmental management. However, uncon-

trolled or poorly planned urban planning can result in sprawl or decay, both associated with their own pitfall and consequences. It used to be that urban growth monitoring relied on manual interpretation of aerial photographs or census data, but these approaches are often too infrequent or subjective. The advent of satellite imagery and advances in machine learning now enable continuous, scalable, and objective monitoring of urbanization patterns. But still most work in this domain focuses on binary classification (urban vs non urban) or land use segmentation at a single point in time, but understanding transitions rather than static states can yield more dynamic insights into the drivers of urban change.

The SpaceNet-7 competition [1] (run by the open satellite imagery dataset project SpaceNet) introduced a large-scale benchmark dataset designed to measure building footprint change detection across more than 100 urban and peri-urban areas worldwide. Although the primary competition objective involved this urban expansion tracking, the imagery's temporal richness also enables exploration of other tasks like as seasonal classification, biome differentiation, and land-use segmentation.

This project uses the SpaceNet-7 dataset to deviate from the competition goal to investigate two alternative problems:

- Season classification from monthly imagery, using both supervised and unsupervised approaches.
- Urban-sector segmentation, distinguishing man-made structures from natural environments and identifying sub-sectors within built regions through Gaussian mixture modeling and engineered features.

A substantial challenge lies in the diversity of regions sampled across the dataset for the rel-

ative sample size. AOIs originate from a wide geographic distribution, encompassing a variety of climates, latitudes, rainfall regimes, vegetation cycles, and developmental patterns. This study therefore serves as an exploration of what can be learned from such imagery, and more in-depth analyses can be extended with more data.

2 Related Work

Previous work on urban classification from satellite imagery has largely focused on static land use mapping or change detection between two discrete timestamps. For example, there are lots of studies and Kaggle projects using satellite imagery to apply Random Forests, CNNs, etc. to differentiate between urban vs not urban, melting ice caps, build up from vegetated areas, deforestation or spread of certain tree species, etc. There is overlap in the initial stages of this project, in terms of creating aforementioned building counters and classifiers and numerical representations of change.

In terms of specific use case overlaps:

2.1 Multi-Temporal Urban Development (SpaceNet-7)

The SpaceNet-7 competition released a globally distributed corpus of monthly satellite imagery with corresponding building-footprint labels. Models submitted to the challenge commonly relied on U-Net or similar architectures to perform instance segmentation or temporal alignment of building masks.

2.2 Seasonality via Machine Learning

Most seasonality detection work focuses on NDVI curves, MODIS time series, or dense temporal products. Rarely is seasonality inferred from only 24 images per AOI, limiting the extendability of the result, but also providing insights into the ability of pure RGB band values to make seasonal inferences.

2.3 Unsupervised Urban-Sector Segmentation

Gaussian Mixture Models (GMMs), as well as edge-aware and texture-based descriptors, are commonly used in preliminary land-use segmentation. While modern approaches often involve semantic segmentation networks, feature-engineered unsupervised models provide interpretability and require minimal ground truth (as

is the case in this project where land use areas are not defined by the project).

3 Materials & Methods

3.1 Data Acquisition and Preprocessing

3.1.1 Imagery and Labels

For each of 60 AOIs, the dataset provides two years of:

- Monthly 1024×1024 RGB GeoTIFF imagery
- Building-footprint labels in GeoJSON format, including a “main” label file for each timestamp and a “match” file linking buildings across time

Imagery was either ingested using Rasterio to preserve geospatial metadata and extract spectral bands or ResNet-18 to extract deep feature embedding.

3.1.2 Image Rescaling and Flattening

Images were resized, either through raw image resizing (e.g. 1024×1024 to 256×256), to reduce runtime and memory footprint. For classical models, pixels were flattened into vectors of raw spectral values.

3.1.3 Dimensionality Reduction

Two parallel pipelines were constructed:

1. Incremental PCA

- Flattened pixel vectors → PCA → 96.9% variance explained
- Produces per image representations that are low dimensional, but have individual red, green, and blue bands which help with seasonal feature extraction.

2. ResNet-18 Embeddings [2]

- Pretrained CNN extracts 512-dimensional feature embeddings.
- Embeddings further reduced via PCA to 75 principal components retaining 90% variance.

3.2 Supervised Season Classification

Seasons were assigned via standard meteorological groupings: Winter: Dec–Feb; Spring: Mar–May; Summer: Jun–Aug; Fall: Sep–Nov. A suite of classifiers was evaluated as displayed in Table X. Inputs were PCA-reduced RGB vectors.

3.2.1 Feature Engineering for AOI-level Seasonality

Further examination into per-AOI season classifiers revealed a wide variation in the model accuracies between AOIs, i.e. certain areas had seasonal patterns that were much easier to model and predict than others. This motivated investigation into which AOI image components are correlated with high modeling success. Per-AOI meta-features were computed using both the original RGB band arrays and the 75 PCA components from the ResNet pipeline, due to the fact that patterns in the red, green, blue, and combined flattened array values for the different seasons were not able to be determined:

- Mean, standard deviation, range for each component
- Total variation
- Total green band variation
- Seasonal contrast value for each season pairing (distance between fall and spring centroids, fall and winter centroids, ...)

The resulting two data frames, the RGB and ResNet-18 feature vectors, served as inputs for a Random Forest Regressor predicting season-classification accuracy per AOI, exploring whether biome/region dependent factors determine model difficulty, or the degree to which the area exhibits predictable seasonal patterns.

3.2.2 Unsupervised Seasonality Modeling

Given the limitations of hard month to season encoding across diverse climates, a Gaussian Hidden Markov Model (HMM) with two latent states was applied to capture whether the AOI exhibits meaningful seasonal fluctuation or not. Additionally, AOI 0331E-1257N-1327 in Contra Costa, California area was identified to have sharply defined seasonal patterns that align with the previous definition. Thus it was fit to a Gaussian HMM with four latent states to observe if the seasonal states cluster together when the region does exhibit meaningful seasonal fluctuation.

3.3 Urban-Sector Segmentation via Gaussian Mixture Models

The image of area 0487E-1246N-1950 taken in June 2019 was identified as a good candidate for land use segmentation as image showed distinct areas of lush vegetation, low lying scattering of homes, and higher density / industrial looking areas.

The general process involved defining features suspected to be indicative of land use for each small patch of the image, then fit to a GMM model where k is the number of land use categories.

Segmentation proceeded in three stages.

3.3.1 Stage 1: Man-Made vs. Natural Mask

For each 8×8 patch:

- Mean and standard deviation per color band
- Edge density (using Sobel library)

Features were clustered using a 2-component GMM, producing a binary urban/non-urban mask.

3.3.2 Stage 2: Density Mask (Within Urban Regions)

Additional features:

- Brightness
- Greenness
- Contrast & homogeneity from skimage.feature library

A 3-component GMM differentiated dense urban cores from sparse building distributions.

3.3.3 Stage 3: Enhanced Sector Mask

To correct color and shape overweighting:

- Corner density
- Number of connected components
- Additional textural metrics (?)

A further 3-component GMM produced enhanced land-use clusters.

Classifier	Accuracy	F1-Score
Bagged RF	0.480702	0.472910
LDA RF	0.473684	0.469493
Random Forest	0.466667	0.464987
KNN	0.466667	0.457261
XGBoost	0.449123	0.447937
SVM	0.435088	0.427811
Naive Bayes	0.266667	0.242460

Table 1: Season Classifier Models Metrics

3.4 Preliminary Time-Series Experiments

Additional exploratory tasks included:

- ARIMA/SARIMA forecasting on per-AOI monthly green-band means, stitching data from the 15 AOIs with the best seasonal predictor model accuracies as if they were 30 consecutive years.
- Random Forest Regression of building count, using GeoJSON footprint lengths as labels.

4 Results

4.1 Season Classification Performance

Most classifiers achieved similar accuracies, with the exception of Naïve Bayes, which underperformed due to feature dependencies. AOI-level variability was substantial: Some AOIs produced near-perfect predictions. Others showed accuracy barely above chance. No clear correlation emerged between AOI performance and spectral band distributions.

4.2 Random Forest Regressor on AOI Meta-Features

A regressor to predict the season model accuracy per AOI was performed, meaning a good regressor model performance would indicate a strong link between the image features and the level of seasonal manifestation of the different AOIs. This model was not successful; two versions were ran with RGB band and ResNet18 image features, and received R^2 scores of -0.03912 and -0.03611 respectively. SHAP feature importance plotting indicates some non-trivial feature values, but only to a max SHAP value of 0.04 (on a scale of the target variable model accuracies, 0 to 1).

This model was first in which the ResNet18 embeddings were deployed, and due to the relative

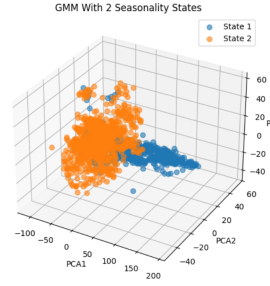


Figure 1: GMM With 2 Seasonality States

Silhouette Score	Davies Bouldin Score	Calinski Harabasz Score
0.2277	1.5586	513.6124

Table 2: Caption

lack of model improvement, they were not utilized for the remainder of the project.

4.3 Biome-Dependent Effects

This result indicated that direct relations between numerical image features and AOI region seasonal prevalence are weak, at least for this specific and rather small dataset.

Example: An AOI appeared to display greater vegetation vibrancy in winter, at first thought to be a labeling error. However the .geojson provided coordinate values (-13.52, -71.97) were found to be in Cuzco, Peru, which experiences dry and wet seasons as opposed to conventional North American ones.

It's possible that regions with non-traditional seasonal patterns, e.g. tropical wet/dry regimes, producing reversed or ambiguous visual cues, are impactful to the analysis especially considering the 60 AOIs featured in this dataset span from around the globe.

4.4 Unsupervised Latent-Season Modeling

Since the regions featured in the dataset are vast in their biome types, it's possible that the hard coding of seasons to 4 discrete months that align with the western calendar is not the ideal approach. But if that is not the right way to group regions, what is? We can approach this unknown by applying latent modeling, specifically a two state Gaussian HMM where each state would refer to one region type, i.e. "seasonal" or "not seasonal". The resulting model groups the states reasonably effectively:

As for the 4 state HMM model applied to just AOI 0331E-1257N-1327, distinct latent states with traditional seasonality were successfully identified. Even though the states are not de-



Figure 2: Unsupervised Seasonal States Over Time (HMM)



Figure 3: Image 0487E-1246N-1950, June 2019

fined (i.e. the model doesn't tell us state $X = \text{winter}$), states do form cycles that cluster around seasonal months.

4.5 Urban-Sector Segmentation

Stage 1:

The 2-component GMM effectively separated man-made and natural patches, especially where structures exhibited high reflectance.

Stage 2:

Three-level clustering differentiated:

Stage 3:

Incorporating corner density and connectivity improved robustness, reducing false positives in vegetated regions.

All Metrics:

4.6 Time-Series Forecasting and Building Regression

ARIMA and SARIMA performance was generally unstable due, likely due to the data being technically synthetic (many AOIS stitched together). SARIMA performance tracked closer to real values.

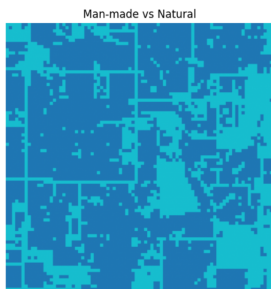


Figure 4: Stage 1 2-component GMM

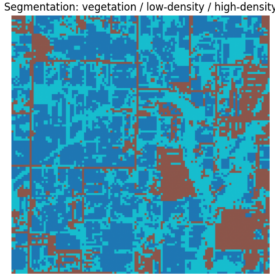


Figure 5: Stage 2 3-component GMM

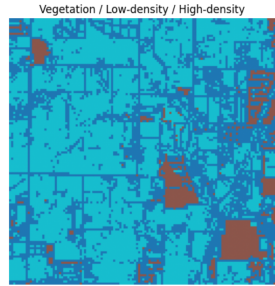


Figure 6: Stage 3 3 component GMM

Stage	Silhouette Score	Davies Bouldin Score	Calinski Harab
1	0.497218	0.911633	6224.75
2	0.235813	1.317310	6955.77
3	0.290023	1.349883	5605.14

Table 3: Caption

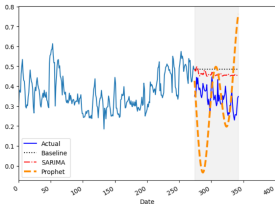


Figure 7: ARIMA/SARIMA results on stacked time values

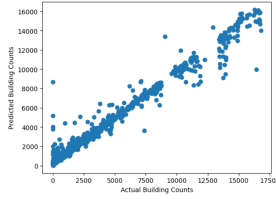


Figure 8: Predicted vs. Actual Values

Building count regression showed initial promise on select AOIs early in the project, and expanding model to full dataset shows decent performance.

Discussion and Conclusions

This study highlights both the potential and the limitations of applying machine learning methods to sparse multi-temporal satellite imagery.

4.7 Key Findings

- Season classification is viable but highly region-dependent. AOIs with strong seasonal signatures yield accurate models; tropical or arid regions complicate traditional seasonal encodings.
- Unsupervised seasonality benefits from latent modeling. However, HMMs with a higher number of states require stronger temporal signals than what is available from monthly RGB imagery alone.
- Urban-sector segmentation via feature engineered GMMs is effective. Multi-stage GMM pipelines achieve clear separation of man-made and natural areas and can be expanded into more complex land use categories.
- Deep embeddings (ResNet-18) do not outperform raw spectral PCA features, in this case.

4.8 Limitations

- RGB-only imagery restricts access to spectral indices.
- Two-year temporal and 60 area world-wide coverage limits seasonal inference.
- Absence of ground truth for land-use categories constrains evaluation.

4.9 Future Directions

1. Expanding building count regression to predicting YOY change in development instead of individual buildings
2. Increased temporal coverage from longer-term satellite archives.
3. Expanding land use mask methodology to other images and types of land use.
4. Enhanced seasonal modeling, perhaps with more robust latent modeling of biomes, rain patterns, uncategorized effects of things like climate change or extreme weather events.
5. Transfer learning across AOIs, using meta-learned shared seasonal representations.

References

- [1] Adam Van Etten. The spacenet 7 multi-temporal urban development challenge: Dataset release.
- [2] Resnet-18 convolutional neural network - matlab resnet18.